Status	Finished
Started	Saturday, 10 October 2026, 7:18 PM
Completed	Saturday, 10 October 2026, 7:23 PM
Duration	4 mins 31 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It does not work with negative numbers
- c. It can only be applied to large lists
- d. It is slower than linear search for small lists

Question **2**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Determining whether an element is present in a list ✓
- b. Inserting elements into a list
- c. Sorting elements in a list
- d. Deleting elements from a list

Question **3**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Hashing
- b. Binary search
- c. Linear search ✓
- d. Jump search

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list contains duplicate elements
c. When the list is sorted ✓
d. When the list is unsorted

Question **5**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. n-1
c. n ✓
d. log n

Question **6**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. n/2
c. log n
d. n ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Must use a sorted array
b. There must be a mechanism to access middle element directly
c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **8**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search
b. Sequential search
c. Linear search
d. Binary search ✓

Question **9**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By dividing the list length by two ✓
b. By starting from the first element
c. By comparing each element sequentially
d. By using a hash function

Question **10**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(n log n)
b. O(1)
c. O👎 ✓
d. O(log n)

Question **11**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 1000
c. 500
d. 10 ✓

Question **12**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)
b. O(n log n)
c. O(log n) ✓
d. O👎

Question **13**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering
b. Mining
c. Finding
d. Searching ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

Number values should only be present

b. There must be a mechanism to delete and/or insert elements in the list ✓
c. The list must be sorted
d. There should be direct access to the middle element in any sublist

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search divides the list into halves.
c. Linear search requires the list to be sorted.
d. Linear search can be applied to both sorted and unsorted lists. ✓